Student Dropout Prediction

MATH 4322 - Group 4 Final Report (Logistic Regression)

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Spring 2026

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1 Introduction

1.1 Data Description (Thanh Trung Do, Leah Padilla)

The dataset is the **Student Dropout Prediction Dataset** from Kaggle (Ali, 2024), containing 10,000 student records with 18 predictor variables and 1 binary response variable.

Response variable:

- `Dropout`: Binary (0 = retained, 1 = dropped out)

Predictor variables (inputs):

- *Demographic*: Age, Gender (Male/Female), Family_Income, Parental_Education (None, High School, Bachelor, Master, PhD)
- *Academic*: GPA, Semester_GPA, CGPA, Semester (Year 1–4), Department (Science, Arts, Business, CS, Engineering)
- *Behavioral*: Study_Hours_per_Day, Attendance_Rate, Assignment_Delay_Days, Travel_Time_Minutes, Stress_Index (1–10)
- *Socioeconomic*: Internet_Access (Yes/No), Part_Time_Job (Yes/No), Scholarship (Yes/No)

Three variables (`Family_Income`, `Study_Hours_per_Day`, `Stress_Index`) have 500 missing values each (5%), which we imputed with their respective medians.

1.2 Research Question

Which factors affect whether a student drops out, and can we accurately predict student dropout?

The response variable is `Dropout` (binary: 0 or 1), making this a classification problem.

2 Methods

2.1 Logistic Regression (Thanh Trung Do, Leah Padilla)

2.1.1 Why Logistic Regression?

Logistic regression is appropriate because our response variable is binary (Dropout = 0 or 1). As discussed in Lecture 8, logistic regression models the probability that Y belongs to one of two categories, and the output is always between 0 and 1 (unlike linear regression, which can predict values outside this range).

Advantages:

- Provides interpretable coefficients: each $\hat{\beta}_j$ tells us the change in log-odds per unit increase in X_j.
- Outputs probabilities, allowing flexible threshold selection.
- Computationally efficient and well-suited for inference (significance testing of predictors).

Disadvantages:

- Assumes a linear relationship between predictors and log-odds, which may not capture complex nonlinear patterns.
- Can be unstable when classes are well-separated (Lecture 10).
- May underperform compared to tree-based methods when interactions between predictors are important.

2.1.2 Model Formula

The logistic regression model is:

$$\log\left(\frac{P(\text{Dropout} = 1 \mid X)}{1 - P(\text{Dropout} = 1 \mid X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

where the left-hand side is the log-odds (logit), and $X_1, \ldots, X_p$ are the 18 predictor variables. The predicted probability is:

$$P(\text{Dropout} = 1 \mid X) = \frac{\exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}{1 + \exp(\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p)}$$

We estimate $\beta_0, \beta_1, \ldots, \beta_p$ using maximum likelihood. In R: `glm(Dropout ~ ., family = binomial)`.

2.1.3 Thought Process for Model Fitting

After fitting the full model with all 18 predictors, we examine the p-values from the `summary()` output. Predictors with large p-values are not statistically significant and may be excluded. We also use **stepwise selection (AIC)** to find a more parsimonious model and compare it against the full model.

2.1.4 Data Preprocessing

```
library(ggplot2)
library(dplyr)
library(corrplot)

data <- read.csv("student_dropout_dataset_v3.csv")

data$Family_Income[is.na(data$Family_Income)] <- median(data$Family_Income, na.rm = TRUE)
data$Study_Hours_per_Day[is.na(data$Study_Hours_per_Day)] <- median(data$Study_Hours_per_Day, na.rm = TR
data$Stress_Index[is.na(data$Stress_Index)] <- median(data$Stress_Index, na.rm = TRUE)

data$Student_ID <- NULL
```

```r
data$Gender <- as.factor(data$Gender)
data$Internet_Access <- as.factor(data$Internet_Access)
data$Part_Time_Job <- as.factor(data$Part_Time_Job)
data$Scholarship <- as.factor(data$Scholarship)
data$Semester <- as.factor(data$Semester)
data$Department <- as.factor(data$Department)
data$Parental_Education <- as.factor(data$Parental_Education)
data$Dropout <- as.factor(data$Dropout)
```

```r
cat("Dropout distribution:\n")
```

```
## Dropout distribution:
```

```r
table(data$Dropout)
```

```
##
##    0    1
## 7646 2354
```

```r
cat("\nDropout rate:", round(mean(data$Dropout == "1") * 100, 2), "%\n")
```

```
##
## Dropout rate: 23.54 %
```

2.1.5 Train/Test Subdivision (80/20, repeated 10 times)

We randomly subdivide the data into 80% training and 20% testing. We repeat this 10 times using `set.seed()` and record the test prediction error (misclassification rate) for each iteration.

```r
seeds <- 1:10
n <- nrow(data)

resample_results <- data.frame(
  Iteration = integer(),
  Error_Rate = numeric(),
  Accuracy = numeric(),
  Sensitivity = numeric(),
  Specificity = numeric()
)

for (s in seeds) {
  set.seed(s)
  idx <- sample(1:n, 0.8 * n)
  tr <- data[idx, ]
  te <- data[-idx, ]

  model <- glm(Dropout ~ ., data = tr, family = binomial)
  prob <- predict(model, newdata = te, type = "response")
  pred <- ifelse(prob > 0.5, 1, 0)
  pred <- factor(pred, levels = c(0, 1))

  cm <- table(Predicted = pred, Actual = te$Dropout)
  tp <- cm["1", "1"]; tn <- cm["0", "0"]
  fp <- cm["1", "0"]; fn <- cm["0", "1"]

  resample_results <- rbind(resample_results, data.frame(
```

```r
    Iteration = s,
    Error_Rate = 1 - (tp + tn) / sum(cm),
    Accuracy = (tp + tn) / sum(cm),
    Sensitivity = tp / (tp + fn),
    Specificity = tn / (tn + fp)
  ))
}

rownames(resample_results) <- NULL
round(resample_results, 4)
```

```
##    Iteration Error_Rate Accuracy Sensitivity Specificity
## 1          1     0.1785   0.8215      0.4365      0.9329
## 2          2     0.1615   0.8385      0.4361      0.9513
## 3          3     0.1955   0.8045      0.4064      0.9268
## 4          4     0.1955   0.8045      0.3760      0.9413
## 5          5     0.2015   0.7985      0.3616      0.9380
## 6          6     0.1805   0.8195      0.4183      0.9431
## 7          7     0.1875   0.8125      0.4116      0.9394
## 8          8     0.1950   0.8050      0.3958      0.9342
## 9          9     0.1985   0.8015      0.3947      0.9332
## 10        10     0.1885   0.8115      0.3925      0.9335
```

```r
cat("Mean test prediction error across 10 iterations:",
    round(mean(resample_results$Error_Rate), 4), "\n")
```

```
## Mean test prediction error across 10 iterations: 0.1882
```

3 Results (Thanh Trung Do, Leah Padilla)

3.1 Model Summary

We fit the logistic regression on the full dataset to examine which predictors are significant.

```
data_num <- data
data_num$Dropout <- as.numeric(data_num$Dropout) - 1
fit.full <- glm(Dropout ~ ., data = data_num, family = binomial)
summary(fit.full)
```

```
##
## Call:
## glm(formula = Dropout ~ ., family = binomial, data = data_num)
##
## Coefficients:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                    1.990e+00  4.314e-01   4.613 3.97e-06 ***
## Age                            1.031e-02  1.285e-02   0.802   0.4224
## GenderMale                    -6.462e-02  5.491e-02  -1.177   0.2392
## Family_Income                 -2.106e-06  1.370e-06  -1.537   0.1242
## Internet_AccessYes            -3.773e-01  8.029e-02  -4.699 2.61e-06 ***
## Study_Hours_per_Day           -1.536e-02  2.251e-02  -0.682   0.4950
## Attendance_Rate               -3.088e-02  3.447e-03  -8.959  < 2e-16 ***
## Assignment_Delay_Days          1.588e-01  1.995e-02   7.959 1.74e-15 ***
## Travel_Time_Minutes            9.707e-03  2.321e-03   4.182 2.89e-05 ***
## Part_Time_JobYes               2.189e-01  5.576e-02   3.925 8.69e-05 ***
## ScholarshipYes                -2.691e-02  5.767e-02  -0.467   0.6408
## Stress_Index                   1.878e-01  1.766e-02  10.639  < 2e-16 ***
## GPA                           -1.099e+00  9.810e-02 -11.206  < 2e-16 ***
## Semester_GPA                  -8.074e-01  6.388e-01  -1.264   0.2063
## CGPA                           7.857e-01  6.320e-01   1.243   0.2138
## SemesterYear 2                 1.902e-01  7.841e-02   2.426   0.0153 *
## SemesterYear 3                 1.409e-01  7.835e-02   1.799   0.0721 .
## SemesterYear 4                 1.367e-01  7.846e-02   1.743   0.0814 .
## DepartmentBusiness            -3.715e-02  8.652e-02  -0.429   0.6676
## DepartmentCS                  -7.328e-02  8.673e-02  -0.845   0.3982
## DepartmentEngineering         -6.473e-02  8.683e-02  -0.746   0.4560
## DepartmentScience             -5.051e-02  8.618e-02  -0.586   0.5579
## Parental_EducationHigh School  6.205e-02  6.557e-02   0.946   0.3440
## Parental_EducationMaster      -8.483e-02  7.674e-02  -1.105   0.2690
## Parental_EducationNone         9.663e-02  1.270e-01   0.761   0.4466
## Parental_EducationPhD          9.080e-02  1.338e-01   0.678   0.4975
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 10914.4  on 9999  degrees of freedom
## Residual deviance:  8309.5  on 9974  degrees of freedom
## AIC: 8361.5
##
## Number of Fisher Scoring iterations: 5
```

3.2 Stepwise Model Selection

```r
step_model <- step(fit.full, direction = "both", trace = 0)
cat("Full model AIC: ", AIC(fit.full), "\n")
```

```
## Full model AIC:  8361.546
```

```r
cat("Stepwise model AIC:", AIC(step_model), "\n")
```

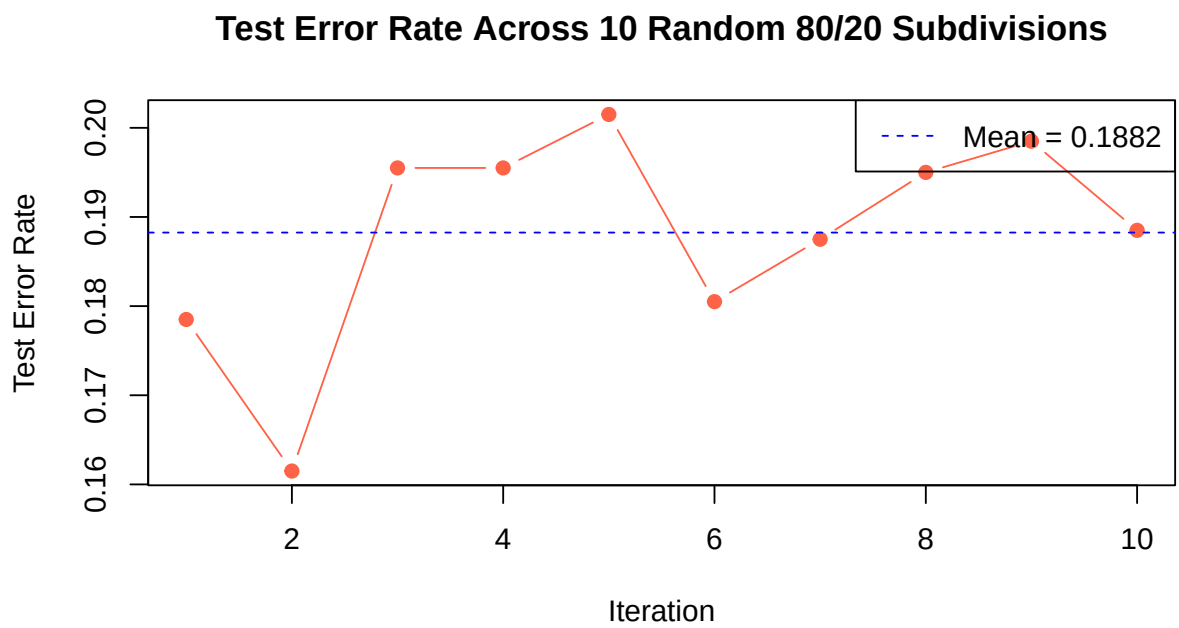
```
## Stepwise model AIC: 8343.138
```

```r
anova(step_model, fit.full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Dropout ~ Family_Income + Internet_Access + Attendance_Rate +
##     Assignment_Delay_Days + Travel_Time_Minutes + Part_Time_Job +
##     Stress_Index + GPA + Semester
## Model 2: Dropout ~ Age + Gender + Family_Income + Internet_Access + Study_Hours_per_Day +
##     Attendance_Rate + Assignment_Delay_Days + Travel_Time_Minutes +
##     Part_Time_Job + Scholarship + Stress_Index + GPA + Semester_GPA +
##     CGPA + Semester + Department + Parental_Education
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      9988     8319.1
## 2      9974     8309.5 14   9.5918   0.7914
```

3.3 Test Prediction Error

```r
plot(resample_results$Iteration, resample_results$Error_Rate,
     type = "b", pch = 19, col = "tomato",
     xlab = "Iteration", ylab = "Test Error Rate",
     main = "Test Error Rate Across 10 Random 80/20 Subdivisions")
abline(h = mean(resample_results$Error_Rate), lty = 2, col = "blue")
legend("topright",
       legend = paste("Mean =", round(mean(resample_results$Error_Rate), 4)),
       lty = 2, col = "blue")
```




```r
cat("=== Logistic Regression - Prediction Error Summary ===\n\n")
```

```
## === Logistic Regression - Prediction Error Summary ===
```

```r
cat("Mean test error (10 iterations):", round(mean(resample_results$Error_Rate), 4), "\n")
```

```
## Mean test error (10 iterations): 0.1882
```

```r
cat("SD of test error:                ", round(sd(resample_results$Error_Rate), 4), "\n")
```

```
## SD of test error:                0.012
```

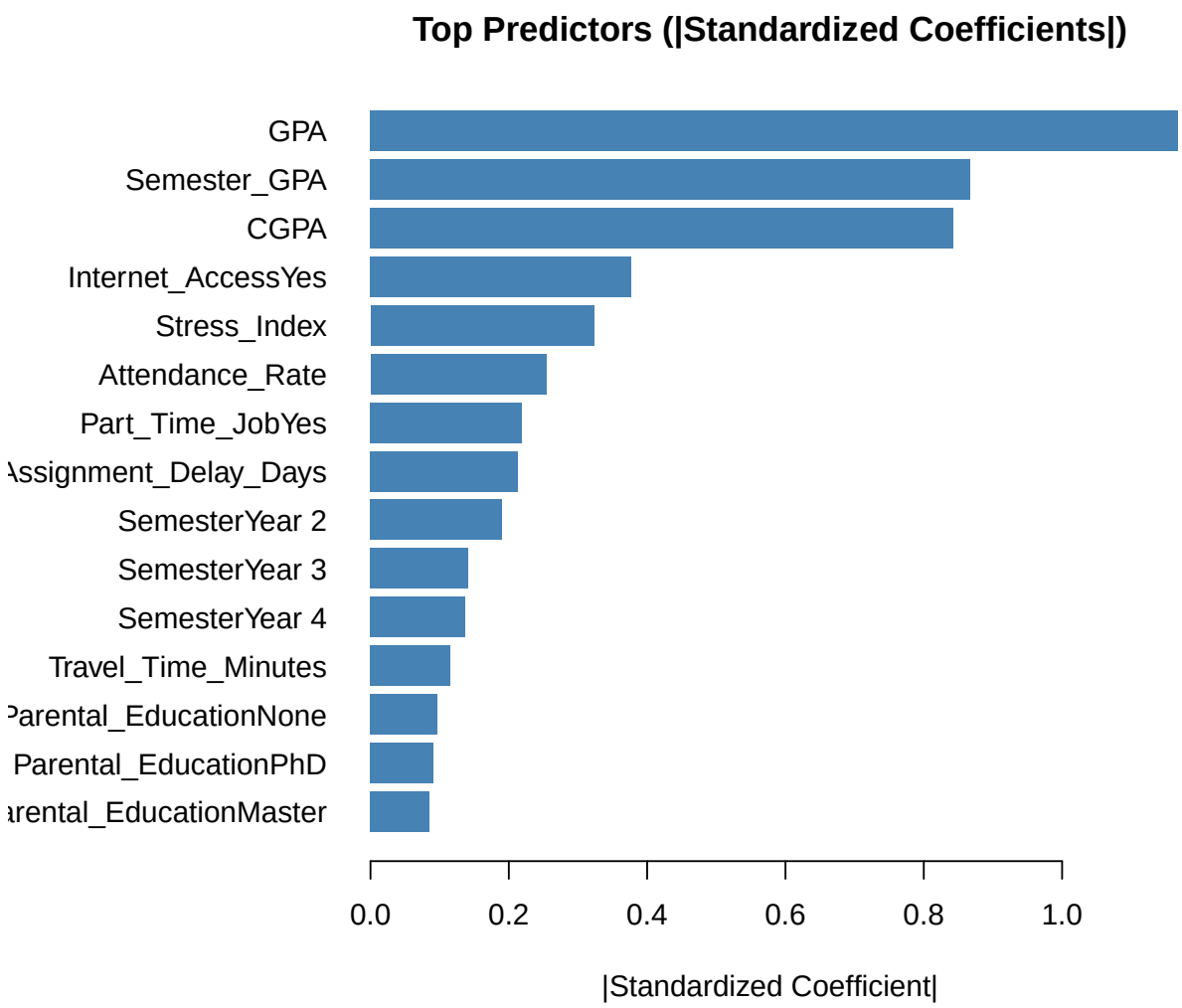
```r
cat("Mean accuracy:                   ", round(mean(resample_results$Accuracy), 4), "\n")
```

```
## Mean accuracy:                   0.8118
```

3.4 Variable Importance

```r
num_vars <- c("Age", "Family_Income", "Study_Hours_per_Day", "Attendance_Rate",
              "Assignment_Delay_Days", "Travel_Time_Minutes", "Stress_Index",
              "GPA", "Semester_GPA", "CGPA")
train_scaled <- data_num
for (col in num_vars) {
  mu <- mean(data_num[[col]]); s <- sd(data_num[[col]])
  train_scaled[[col]] <- (data_num[[col]] - mu) / s
}
model_scaled <- glm(Dropout ~ ., data = train_scaled, family = binomial)
std_coefs <- sort(abs(coef(model_scaled)[-1]), decreasing = TRUE)
top_n <- min(15, length(std_coefs))
par(mar = c(5, 10, 3, 2))
barplot(std_coefs[top_n:1], horiz = TRUE, las = 1,
        col = "steelblue", border = NA,
        main = "Top Predictors (|Standardized Coefficients|)",
```

```
        xlab = "|Standardized Coefficient|")
```



3.5 Interpretation

- **Academic performance** variables (GPA, CGPA, Semester_GPA) are among the strongest predictors. Students with lower GPAs have significantly higher odds of dropping out.
- **Behavioral factors** such as Attendance_Rate and Study_Hours_per_Day are also significant.
- The model achieves a **mean test error rate** of 0.1882 across 10 random 80/20 subdivisions, with low standard deviation (0.012), confirming stable predictive performance.

4 Conclusion

- Logistic regression achieves a mean test prediction error of **0.1882** across 10 random subdivisions (80/20).
- The significant predictors (GPA, Attendance_Rate, Study_Hours_per_Day, etc.) align with domain knowledge about student success factors.
- The low standard deviation across iterations confirms model stability.

Note: The Decision Tree results and final model comparison will be added by the Decision Tree team (Sriharsha, Kelvin, Tien).

5 References

1. Ali, M. (2024). Student Dropout Prediction Dataset. Kaggle. https://www.kaggle.com/datasets/meharshanali/student-dropout-prediction-dataset
2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning with Applications in R* (2nd ed.). Springer.
3. R Core Team (2025). R: A Language and Environment for Statistical Computing. https://www.R-project.org/

6 Appendix: Full R Source Code

The full R source code is attached separately.